

## SMART INDIA HACKATHON 2026

# SthiraRoute

Sthira (■■■■■) — steady. A fleet plan that survives the day.

|                   |                                                                    |
|-------------------|--------------------------------------------------------------------|
| Team              | HackShastra                                                        |
| Team Leader       | Santosh Maurya                                                     |
| Roll Number       | B25BS1530                                                          |
| Domain            | Software                                                           |
| Problem Statement | PS-02 — Smart Fleet Coordination and Logistics Management Platform |
| Theme             | Transportation & Logistics                                         |

## Production deployment

<https://rohensingh.in>

## Source repository

<https://github.com/BlaxeZ/SthiraRoute>

One board plans the shift, proves the load physically fits, and keeps the plan alive when the day breaks — measured on Bengaluru, Mumbai, Delhi NCR and Hyderabad with real road distances.

## 01 · Deployment & access

SthiraRoute is deployed as a production web application. The public demo and source repository are listed below.

| Item             | Detail                                                                                    |
|------------------|-------------------------------------------------------------------------------------------|
| Live application | <a href="https://rohensingh.in">https://rohensingh.in</a>                                 |
| Source code      | <a href="https://github.com/BlaxeZ/SthiraRoute">https://github.com/BlaxeZ/SthiraRoute</a> |
| API health       | <a href="https://rohensingh.in/health">https://rohensingh.in/health</a>                   |
| Runtime stack    | FastAPI · Google OR-Tools · OSRM · React · Vite · MapLibre · nginx · systemd              |
| Hosting          | Ubuntu 24.04 LTS VPS (India) · TLS via Let's Encrypt · domain rohensingh.in               |
| Data / cities    | Bengaluru · Mumbai · Delhi NCR · Hyderabad city packs with seeded fleet & demand          |
| Access mode      | Open demo — no login required                                                             |

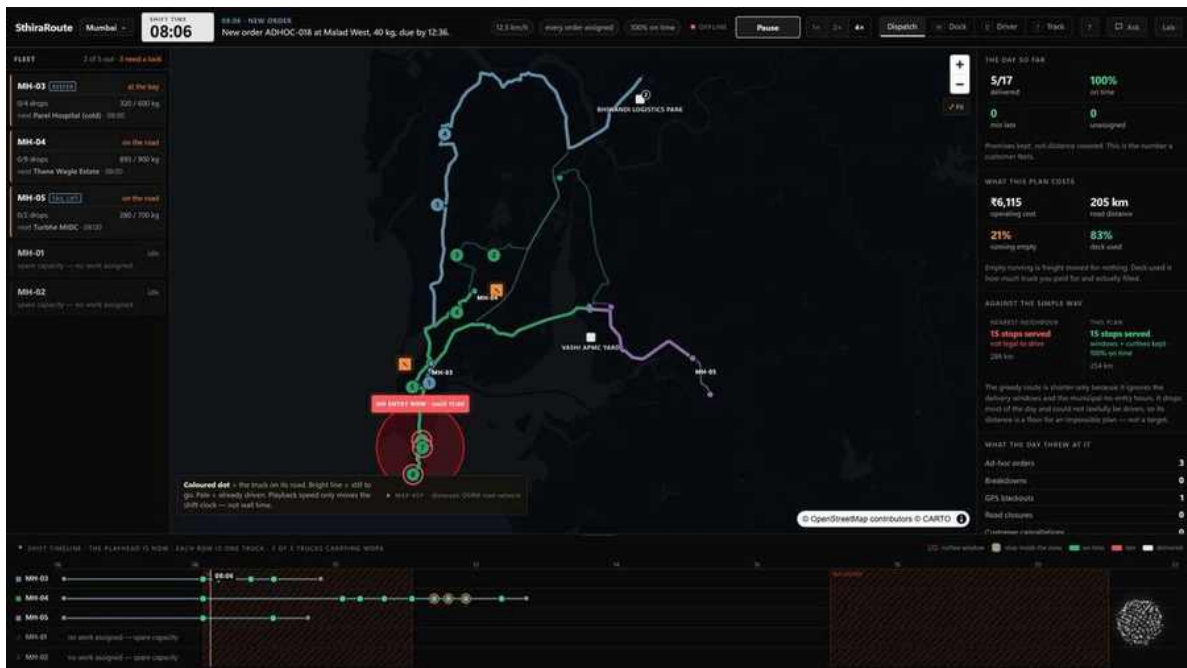
## Product capabilities mapped to PS-02

| Requirement                | SthiraRoute delivery                                             |
|----------------------------|------------------------------------------------------------------|
| Real-time fleet monitoring | Live map, fleet rail, shift clock, WebSocket events              |
| Route optimisation         | OR-Tools CVRPTW with stability, cost, SLA and curfew constraints |
| Load allocation            | Weight/volume capacity + 3D LIFO deck packing with axle check    |
| Shipment tracking          | Track desk, driver manifest, proof of delivery                   |
| Stakeholder coordination   | Shared committed plan across Dispatch, Dock, Driver, Track       |

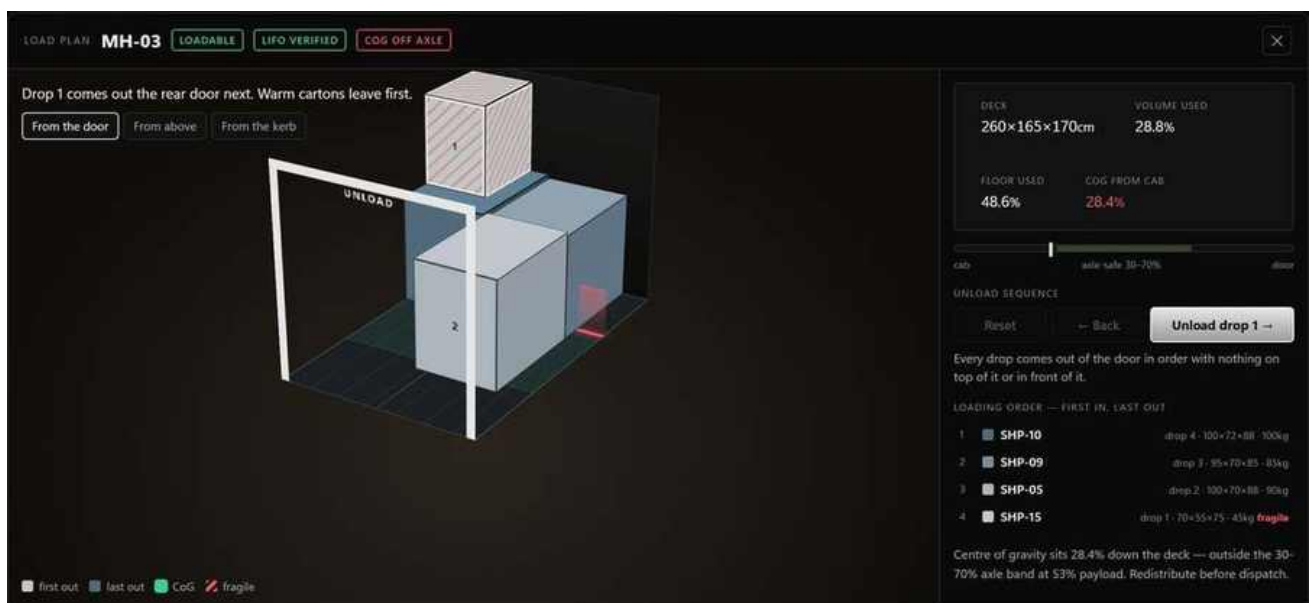
## Operating path on the live system

- 1 Prepare shift (**P**) — plan every order, pack decks, dispatch the fleet.
- 2 Run the day (**Space**) — trucks follow road polylines under the shift clock.
- 3 Inspect a route load plan — LIFO unload, fragility, support, centre of gravity.
- 4 Lab (**L**) — breakdown, ad-hoc order, road closure; stability-aware re-plan.
- 5 Desks — Dock / Driver / Track read the same committed plan.

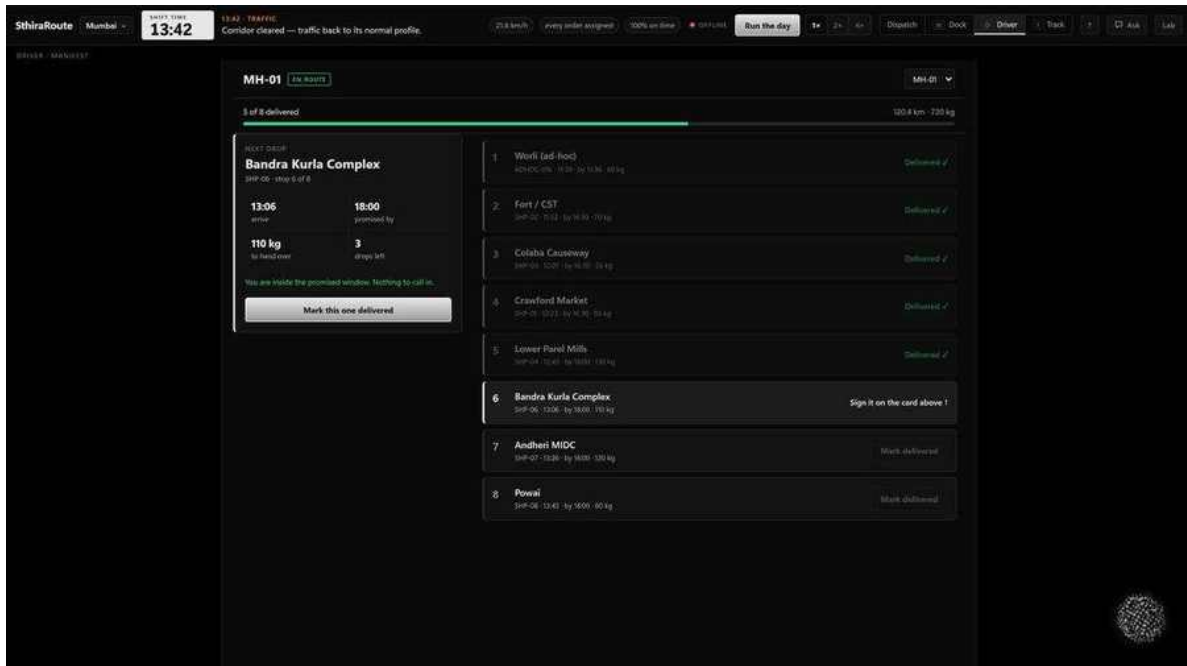
## 02 · Product screenshots



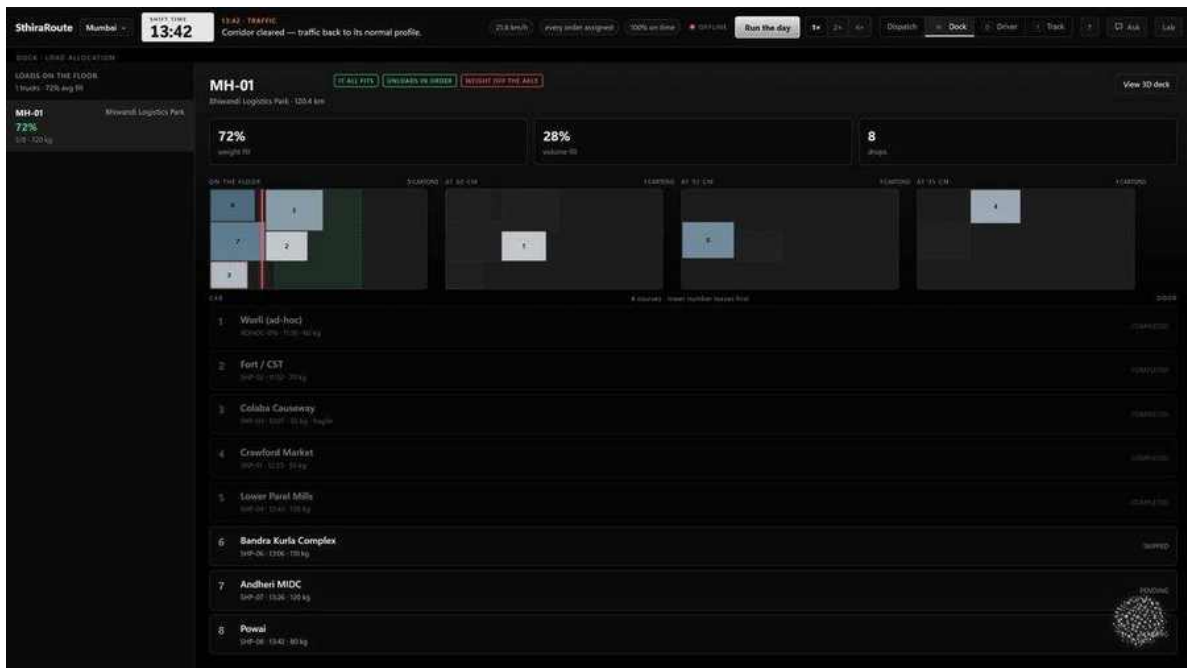
Dispatcher command board — map, fleet, KPIs, municipal no-entry, shift timeline



3D load plan — LIFO verification, fragility, axle / centre-of-gravity band



Driver desk — committed manifest and proof of delivery



Dock desk — bay loading against the committed plan

The screenshot displays the SthiraRoute Track desk interface. At the top, it shows the location 'Mumbai' and a 'SWITCH TIME' of '13:42'. A status bar indicates 'TRAFFIC: Corridor cleared — traffic back to its normal profile.' and '25.8 km/h'. Below this, a 'Run the day' button is visible. The main section is titled 'SHP-01' and 'Crawford Market'. It shows a timeline of events for the consignment, including 'Pending', 'Assigned', 'In transit', and 'Delivered'. A table lists various consignment items with their status (e.g., 'SHP-01 Crawford Market', 'SHP-02 Fort / CST', 'SHP-03 Colaba Causeway'). A detailed log on the right side provides a chronological record of events, such as 'Min-01 has lost GPS. Showing last-known position with an uncertainty ring — the plan is unchanged.' and 'Min-01 delivered SHP-04 on time, 6 min at the bay.'.

Track desk — consignment visibility across the network

### 03 · Idea presentation

Official SIH idea deck for HackShastra — SthiraRoute (Problem Statement 2).

# SMART INDIA HACKATHON 2026



## SthiraRoute

*Sthira (स्थिर) — steady. A fleet plan that survives the day.*

Problem Statement ID —

**2**

Problem Statement Title—

### Smart Fleet Coordination and Logistics Management Platform

Theme—

**Transportation & Logistics**

PS Category—

**Software**

Team ID—

Team Name (Registered on portal)

**Hack Shastra**

One board plans the shift, proves the load physically fits, and keeps the plan alive when the day breaks.

Measured on four city packs — Bengaluru, Mumbai, Delhi NCR and Hyderabad — over real OpenStreetMap road distances.

Route optimisation

3D load planning

Live tracking & re-planning





# SthiraRoute — Smart Fleet Coordination Platform



## One board runs the whole shift

Plan every order onto the fleet · prove the load fits · dispatch to four desks · absorb breakdowns, rush orders and curfews — with every trade-off priced in rupees and the dispatcher always in charge.

### Four failures in the problem statement, four mechanisms

red is the problem, green is what we built



#### Vehicles run half empty



Weight, volume and deck geometry per truck, plus ₹800 for every truck used — so the fleet consolidates itself.

Bengaluru: 15 orders on 2 trucks, 92% payload used



#### Routes planned by hand



OR-Tools over real OSRM road matrices, with delivery windows, curfews, cold-chain and two depots all binding the model.

10–31% less road distance, and legal to drive



#### No live visibility



Trucks move along the very polyline the map draws, with per-stop ETA, proof of delivery and a widening ring when GPS drops.

20 GPS blackouts survived, no freight lost



#### Everyone on a different page

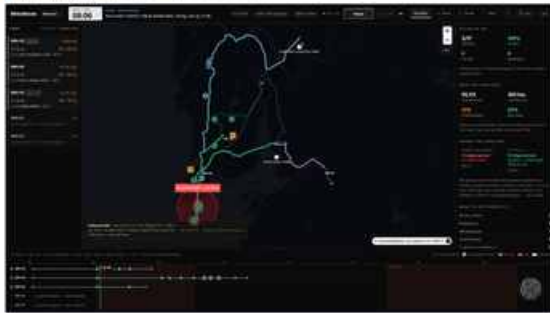


One committed plan read from four desks — Dispatch, Dock, Driver, Track — plus an assistant that explains it in plain words.

Same plan object, four views, one source of truth

### Working prototype

Mumbai, 08.06



### Why this is not another routing demo



#### Churn has a price, so plans stay steady

Moving freight off the truck it was promised to costs ₹150 inside the objective — a re-plan is only offered when the gain beats the disruption.



#### Every route is checked against physics

A 3D packer proves the deck loadable before a truck moves: LIFO order, nothing floating, nothing stacked on fragile, centre of gravity in band.



#### Routes obey the city, by construction

A no-entry window is time and geography: arrival times inside a banned zone are removed from the model, so the truck waits it out.



#### It never just answers INFEASIBLE

Every order carries a drop penalty, so a plan always returns — with a diagnosed reason and a relaxation menu for anything it could not solve.





# TECHNICAL APPROACH



**Technology stack** · open source end to end — no paid routing key, no GPU, no licence



**How a shift runs** · plan across the top, recover along the bottom — one closed loop



**System architecture** · four desks, one API, one committed plan — and the load proved before a truck moves





## FEASIBILITY AND VIABILITY



### 191

#### automated checks pass

53 pytest tests in 70 s · 93 interface checks in headless Chromium · 45 end-to-end assertions

### 10–31%

#### less road distance

measured in all four cities against a greedy baseline on the same OSRM roads

### 4

#### city packs, not one skin

each with its own depots, curfew windows, hazard corridors and congestion curve

### ₹0

#### licence cost to run

OR-Tools, OSRM, MapLibre, FastAPI, PostgreSQL — open source throughout

**The growth path is a config file, not a rewrite** — the solver never changes



#### Today — one laptop

##### BUILT AND MEASURED

SQLite, the solver in-process, the public OSRM. No Docker, no cluster, no map key. A whole shift plans in an 8-second solver budget.



#### Pilot — one VM

##### ONE DEPOT, ONE CITY, FOUR WEEKS

PostgreSQL + PostGIS and a self-hosted OSRM for one state. Runs in shadow beside the whiteboard while km and on-time rate are compared daily.



#### Scale — add a worker

##### CPU ONLY, NO PER-REQUEST FEE

Redis-backed workers take the solve off the request path. The recurring bill stays one server and one database — which is what mid-size fleets can afford.

**Risks we can name, and what we already do about them**



#### The public OSRM server is shared and rate-limited

Matrices and geometry are LRU-cached, and the system degrades to a Haversine fallback at a ×1.4 defour factor with the source labelled in the UI. A pilot self-hosts OSRM.



#### OpenStreetMap truck restrictions are thin in India

OSRM stays for geometry and matrices; Mappls vehicle-class routing supplies restriction-accurate distances the moment a pilot needs them. The solver does not care which provider answers.



#### A live re-plan cannot take minutes

Re-planning warm-starts from the plan already on the road and runs inside a hard time budget — four seconds for a disruption. A ₹150 churn price stops it reshuffling the day.



#### Curfew windows here are templates, not gazetted law

Every zone is per-city configuration meant to be sourced from traffic-police notification, and every seeded coordinate and window is labelled an assumption in the source.



#### Dispatchers do not trust a box that reshuffles their day

Recommend-and-approve: a re-plan lands as a proposal, drawn dashed on the map, showing what it saves, what it costs and how much freight it moves before anyone clicks.



#### A language model could invent numbers, or move freight

Gemini is read-only over a compact snapshot and answers from tool calls. It can point at the board and explain it, there is no code path by which it can alter a plan.



## IMPACT AND BENEFITS



### MEASURED, NOT PROJECTED

One Mumbai shift: 15 orders, 3 trucks used of 5 · real OSRM road distances

**254 km**

road distance planned  
greedy baseline: 288 km

**100%**

on time, 0 unassigned  
every order served

**18.4%**

driven empty, exposed  
the number nobody measures

**68%**

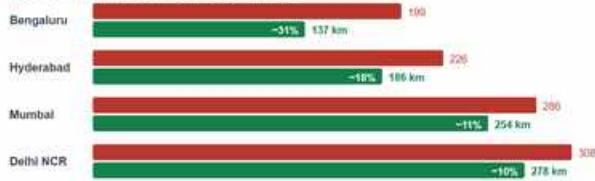
payload capacity used  
weighed, not guessed

**₹7,992**

shift cost, itemised  
every item in rupees

### Distance, city by city · same orders, same roads

■ greedy nearest-neighbour, ignores windows and curfews  
■ SthiraRoute plan — windows and curfews kept, 0 unserved



### A full day in each city, run headless · nothing staged

WHAT THE FOUR DAYS THREW AT IT

- 21 rush orders arrived mid-shift
- 12 orders cancelled after dispatch
- 4 trucks broke down
- 20 GPS blackouts
- 4 municipal closures
- 17 re-plans priced and approved

WHAT CAME OUT THE OTHER SIDE

**69 of 69**  
orders delivered · 0 freight lost

**97%** on time across the four days

**0** orders silently dropped

Mumbai and Bengaluru finished 100% on time. Delhi and Hyderabad each lost a single window.

### Who feels it

**Dispatcher**  
One board instead of a whiteboard and six phone calls

**Dock & warehouse**  
Load order from the packer: first drop nearest the door

**Driver**  
A manifest that matches how the truck is really loaded

**Customer**  
A promised window the plan is built to keep

**Fleet owner & city**  
No-entry hours obeyed, and an audit trail behind every call

### Economic

Ten to thirty per cent fewer kilometres on the same orders is fuel and driver-hours saved every shift, and empty running finally has a number on it.

### Environmental

Fewer kilometres means less diesel and less CO<sub>2</sub>. Trucks wait a curfew out at a bay instead of idling at a cordon, and light days take whole trucks off the road.

### Social & regulatory

No-entry hours are obeyed by construction, not by a driver's judgement at a barricade — so lateness stops being blamed on the person at the wheel.

### Resilience

Four headless days absorbed breakdowns, rush orders, cancellations, GPS blackouts and closures without losing a single consignment.



## RESEARCH AND REFERENCES



### Every claim is traceable

Policy from official sources, method from the optimisation literature, numbers from our own harness on the running system — and where we cannot support a claim, we say we are not making it.

#### Policy & national context

National Logistics Policy, launched 17 Sep 2022 — logistics cost toward single digits, via ULIP and PM GatiShakti.  
[pmindia.gov.in](https://pmindia.gov.in)

NLP three-year update: ULIP now spans 30+ systems and 160+ core transactions (Aug 2025).  
[pib.gov.in](https://pib.gov.in) · [PRID 2167224](https://prid2167224)

ULIP — the interface joining VAHAN, SARATHI, FASTag, e-Way Bill, ICEGATE and state no-entry layers.  
[goupl.in](https://goupl.in)

PM GatiShakti — DPIIT portals and connected initiatives.  
[dpiit.gov.in](https://dpiit.gov.in)

Invest India — the four NLP pillars (IDS, ULIP, ELOG, SIG) beside GatiShakti and Bharatmala.  
[investindia.gov.in](https://investindia.gov.in)

Our four desks map onto roles the policy already names — shipper, carrier, dock, driver — so a pilot needs no new vocabulary.  
**our position**

#### The size of the problem

World Bank: India ranks 36 of 139 on the Logistics Performance Index 2023, the goal is top 25 by 2030.  
[worldbank.org](https://worldbank.org)

NCAER / DPIIT: logistics cost assessed at 7.8–8.9% of GDP for 2021–22 under that methodology.  
[ncaer.org](https://ncaer.org)

The older ~13–14% of GDP narrative sits against that assessment because the two methods differ.  
[pmindia.gov.in](https://pmindia.gov.in) · [ncaer.org](https://ncaer.org)

So our KPIs are deliberately *feet-operational* — km, on-time rate, empty running, payload used.  
**our position**

Every rupee in the objective is a configurable cost model, tunable to an operator's economics.  
**our position**

Our benchmark is like-for-like on one shift, on one road network — never extrapolated to a fleet-year.  
**our position**

#### Optimisation research

Google OR-Tools — vehicle routing framework, distance matrices and constrained variants.  
[developers.google.com/optimization](https://developers.google.com/optimization)

OR-Tools routing options — Guided Local Search documented as generally the most efficient metaheuristic here.  
[developers.google.com](https://developers.google.com)

Gmira et al., *Transportation Research Part C*, 2021 — dynamic VRP with parallel ALNS re-optimisation.  
[sciencedirect.com](https://sciencedirect.com)

Ammounova et al., *Applied Sciences*, 2022 — simheuristics for VRP under uncertainty.  
[mdpi.com](https://mdpi.com)

Tresco et al., *IEEE* 2024 — matheuristic delivery planning for the dynamic VRP.  
[ieeexplore.ieee.org](https://ieeexplore.ieee.org)

3L-CVRP — the field's warning that a geometric packer inside every route move is intractable, which is why we pack once per settled route.  
**our reading of the field**

#### Engines, maps & market

OSRM vs Valhalla vs GraphHopper — routing engines compared on algorithms and deployment.  
[maki.dev](https://maki.dev) · [geofabrik.de](https://geofabrik.de)

OSM truck-restriction completeness is a known gap for commercial vehicle routing in India.  
[help.openstreetmap.org](https://help.openstreetmap.org)

Mappis / MapmyIndia — vehicle-category routing APIs, heavy vehicles included, for Indian roads.  
[developer.mappis.com](https://developer.mappis.com)

MapmyIndia InTouch and Mappis IoT — telematics platforms already operating in this market.  
[intouch.mapmyindia.com](https://intouch.mapmyindia.com)

Locus and FarEye — incumbents. Our gap is affordable mid-fleet optimisation with stable re-planning.  
[locus.sh](https://locus.sh)

MapLibre GL with OpenStreetMap tiles — no per-map-load billing, unlike the proprietary map SDKs.  
[maplibre.org](https://maplibre.org)

#### What we measured ourselves, rather than cited

**53 pytest tests pass in 70 seconds** — 3D packing and LIFO (16), the routing objective (14), the shift simulator (14), the HTTP API (7) and the assistant (2). Alongside them, **93 interface checks** run in headless Chromium and **45 assertions** run end to end against the live API.

#### How the numbers on these slides were produced

A single script seeds each city, plans the shift, records the KPIs and the greedy baseline, then runs the whole simulated day and prints the scorecard. Every figure here comes from that run, over live OSRM road distances — re-runnable in about four minutes for all four cities.

#### What we explicitly do not claim

That a student platform shifts national logistics cost as a share of GDP. That OpenStreetMap matches Mappis for Indian addresses. That we have done a formal prior-art search. The city packs are India-shaped but synthetic, and curfew windows are plausible templates, not gazetted law.

# HackShastra

SthiraRoute

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<https://rohensingh.in>

<https://github.com/BlaxeZ/SthiraRoute>

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Smart India Hackathon 2026 · Problem Statement 2 · Software